

DENVar: A Global Dynamic Estimate of NDVI Variance

Stefano Mezzini^{1,2}, Gavin L Simpson³, and Michael J Noonan^{1,2,4,*}

¹Okanagan Institute for Biodiversity, Resilience, and Ecosystem Services, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

²Department of Biology, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

³Department of Animal and Veterinary Sciences, Aarhus University, Denmark.

⁴Department of Computer Science, Math, Physics, and Statistics, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

*Address correspondence to: michael.noonan@ubc.ca

Abstract

Remote-sensing proxies of habitat productivity such as the Normalized Difference Vegetation Index (NDVI) have helped develop and support many hypotheses and projects. However, historically, users tended to focus solely on trends in mean productivity. In recent years, the variance around the mean (i.e., stochasticity) has gained more attention, particularly as climate change and human-induced rapid environmental change transform the ecosystems species have evolved in and adapted to. Still, calculating trends in NDVI stochasticity requires careful modeling, since stochasticity estimates are conditional on the estimated mean and the scale at which the data are collected and modeled. We thus present a new global Dynamic Estimate of NDVI Variance, *DENVar*. We estimate spatiotemporal trends in NDVI mean and variance using Generalized Additive Models, which provide a flexible yet transparent and interpretable model structure. We show that *DENVar* varies nonlinearly with **XXX** and can be used to test many hypotheses related to environmental stochasticity, such as optimal foraging theory and regime shifts as well as for conservation. We provide a publicly and freely available repository for downloading rasters of daily and long-term averages of *DENVar* and the related estimates of mean NDVI.

1 Introduction

The Normalized Difference Vegetation Index (NDVI) is a proxy of primary productivity used in a wide range of disciplines, including ecology (Pettorelli et al., 2005, 2011), landscape phenology (Merkle et al., 2016; Hmimina et al., 2013), agriculture (Franch et al., 2017), forestry (João et al., 2018), and conservation (Mallegowda et al., 2015; Marcus et al., 2025). High-resolution, remotely-sensed NDVI data have allowed analysts to estimate plant biomass without fieldwork (Bhatti et al., 2024; Borowik et al., 2013) and leverage such estimates for studying large-scale plant dynamics (Pettorelli et al., 2005) and their effects on animal movement behavior (Merkle et al., 2016; Aikens et al., 2017). Still, most NDVI work is limited to quantifying changes in and responses to mean NDVI while ignoring changes in the variance around the mean.

Environmental stochasticity has been studied for decades (Levins, 1974; Stephens and Charnov, 1982; Lande, 1993; Sæther, 1997), but recent work has drawn more attention to the effects environmental stochasticity has at a wide range of scales, from individual organisms (Mezzini et al., 2025; Rizzuto et al., 2021) to landscapes (Zhang et al., 2009; Mutti et al., 2020; Marcus et al., 2025). However, estimating trends in NDVI variance requires more nuanced modeling than estimating changes in mean NDVI, particularly when environments are temporally dynamic and spatially heterogeneous. The scale at which processes are observed determine how individuals’ perception of and ability to respond to change (Levin, 1992). A forest fire is severely disruptive and deadly for many insects (Fettig et al., 2022), while birds and large mammal may not struggle to escape the fire and or establish a new range (Franklin et al., 2021; Calhoun et al., 2024). Similarly, the effects of the fire recovery will depend on its duration relative to other organisms’ phenology and lifespan (Frankham and Brook, 2004). Therefore, distinguishing between a change in mean NDVI and a stochastic fluctuation requires a conscious decision about the scale at which to consider the data.

Previous work suggests that environmental stochasticity can reduce limit species’ ability to adapt or specialize (Chevin et al., 2010), alter species’ movement behavior (Mezzini et al., 2025), and even affect extinction risk at both the population and species levels (Lande, 1993; Bull et al., 2007; Fung et al., 2018). Still, the effects of environmental stochasticity on species dynamics and ecosystem stability remain understudied (Marcus and Noonan, 2023). Global, temporally dynamic estimates of NDVI variance would facilitate research by building upon the widespread use and interpretability of NDVI while also removing the barriers of computational demand and statistical knowledge that are required for quantifying environmental stochasticity. However, since NDVI is not a direct measurement primary productivity but a proxy, it should be interpreted carefully (Pettorelli et al., 2011). NDVI is prone to sensitivity loss in very green environments (Matsushita et al., 2007; Son et al., 2014), values vary across sensors (Beck et al., 2011; Fan and Liu, 2016; Huang et al., 2021), and the coarse spatiotemporal scale of satellite-derived NDVI can be a substantial barrier when assessing fine-scale trends (Goodchild and Quattrochi, 2023; Reid et al., 2018). However, careful statistical modeling can help overcome some of these limitations. In particular, Generalized Additive Models (GAMs) are particularly convenient due to their ability to learn flexibly from data (Wood, 2017; Simpson, 2018) and handle very large spatiotemporal datasets (Wood et al., 2015, 2017).

We present a new measure of environmental stochasticity, which we name the Dynamic Estimate

of NDVI Variance (DENVar). We calculate DENVar using 44 years of NDVI data from the AVHRR and VIIRS collection of sensors and GAMs. Although satellites with AVHRR and VIIRS sensors provide data at a coarser spatial resolution (0.05°) than other satellites (e.g., 250m for MODIS), they are the only sensors to provide data at a daily frequency and for more than 40 years. Our use of GAMs provides a flexible yet interpretable approach for estimating trends in mean and variance in NDVI without the rigidity of other more traditional methods (e.g., ARIMA) or the opaqueness of machine learning models such as neural networks (Simpson, 2018; Wood, 2017). We illustrate the relationships between DENVar and multiple, relevant ecological metrics, and we provide access to a free repository of daily, global estimates of DENVar and the corresponding estimates of mean NDVI.

2 Methods

2.1 Choice of color palettes

We represent NDVI using a modified version of Fabio Crameri’s divergent *bukavu* palette (Crameri, 2018a; Crameri et al., 2020), which has high-contrast for deuteranope and protanope vision. For NDVI values between -0.1 and 1, the colors also have sufficient contrast for the colors to also be distinguishable by strong tritanope and achromatic vision. Fig. A.1 contains an approximate representation of how each vision type would perceive each color palettes used. We obtained all color palettes with three or more colors from the *khroma* package (v. 1.14.0 Frerebeau, 2024) for R (v. 4.4.0; R Core Team, 2024).

2.2 Datasets used

Fig. 1 illustrates the processes of data collection, cleaning, and modeling used in this project. We downloaded NDVI image composites directly from the servers for data from AVHRR sensors (1981-06-24 to 2013-12-31; United States National Centers for Environmental Information (NCEI) and United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2018) and VIIRS sensors (2014-01-01 to 2025-06-29; United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2022).

We removed all pixels labeled as "cloudy" for AVHRR rasters and "probably cloudy" or "confidently cloudy" for VIIRS rasters. To remove pixels that included large portions of water, we used a shapefile of major water bodies (Garmin International, Inc. and Esri, 2025) to produce a static raster of the percentage of water within each pixel, at the same spatial resolution as the NDVI rasters (0.05° ; Fig. A.2). We removed all pixels with $> 40\%$ water cover.

Many rasters had horizontal bands of $\text{NDVI} > 0.9$ at unrealistically northern latitudes between September and March, likely due to limited visibility during fall and winter in the Northern hemisphere. We identified and removed problematic pixels using the monthly empirical cumulative distribution functions of NDVI across latitude (Fig. A.3). We did not filter problematic values at latitudes $< 60^\circ\text{S}$ since we fully excluded Antarctica from the analyses (Fig. A.4).

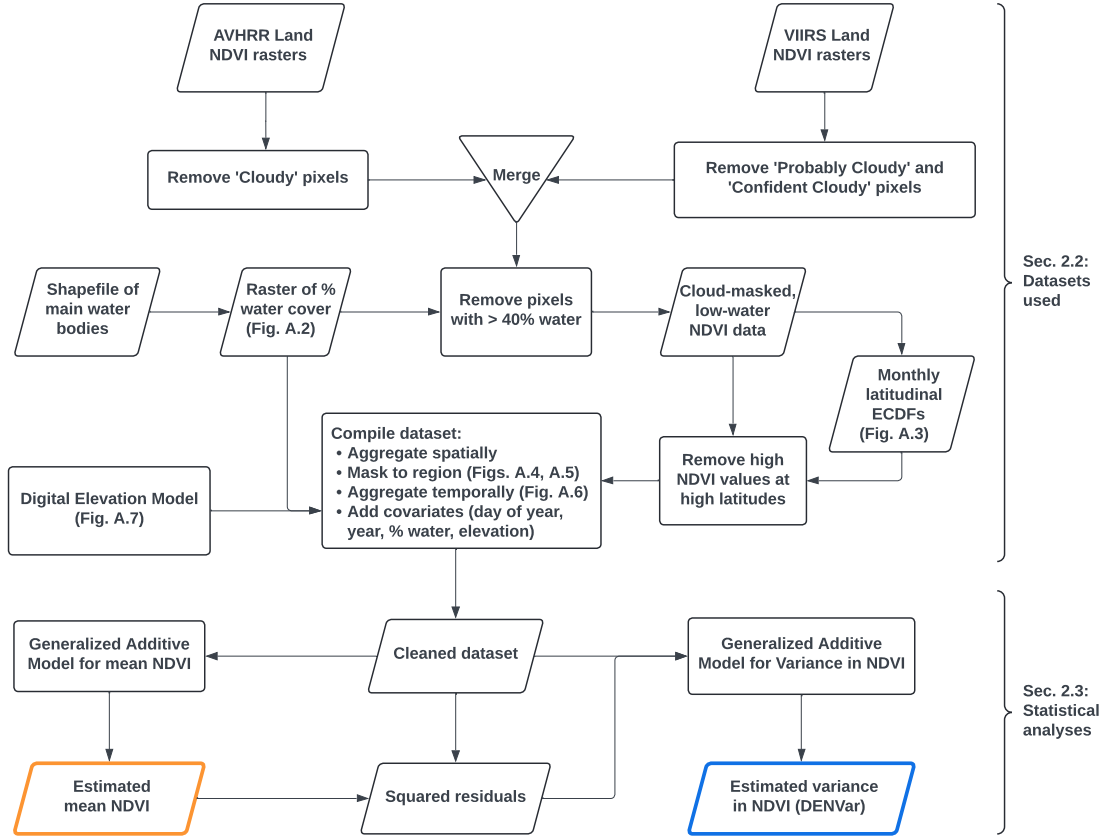


Figure 1: **Flowchart for the analyses used in the manuscript.** The curly braces on the right indicate the subsections in the Methods section that detail the dataset collation and cleaning (2.2), and statistical analyses (2.3).

To ensure the number of rows for each dataset was below the maximum data frame size in R ($3.17 \times 10^{10} > 2^{31} - 1 \approx 0.21 \times 10^{10}$), we reduced the dataset size by splitting it across four spatial groups (Figs. A.4 and A.5) and spatially and temporally aggregated the rasters by a factor of four. The final dataset thus had one raster every four days with a spatial resolution of $0.05^\circ \times 4 = 0.2^\circ$ (Fig. A.6). Finally, we annotated the dataset with columns of day of year (1-366), integer year (1981-2025), percent water cover, and elevation above sea level (m). The digital elevation model was obtained using the `get_elev_raster()` function from the `elevatr` package for R (v. 0.99.0; Hollister et al., 2023) with an original resolution of 0.076° (Fig. A.6). Despite the substantial reduction in dataset size, the dataset creation was still challenging for the machine used, which had 2.2 TB of RAM and two Intel Xeon Platinum 8462Y+ processors (64 cores total).

2.3 Statistical Analyses

We estimated spatiotemporal trends in mean and variance in NDVI using Generalized Additive Models (GAMs) via the `mgcv` package for R (v. 1.9-3 Wood, 2017). To reduce modeling fitting times

with negligible losses to model accuracy, we used the `bam()` function with fast REstricted Marginal Likelihood (`method = fREML`) and covariate discretization (`discrete = TRUE`; Wood et al., 2015, 2017). Ideally, NDVI should be modeled using beta location-scale models (after the linear transformation $Y^* = (Y + 1)/2$) to account for the fact that: (1) NDVI is bounded between -1 and 1, and (2) the variance in NDVI is dependent on the mean (and vice-versa), since ecosystems with very low NDVI (e.g., rock, ice, or concrete) or very high NDVI (e.g., dense forest) tend to have lower variance in NDVI. However, preliminary tests and work by (Marcus et al., 2025) showed that beta GAMs and particularly beta location-scale GAMs had prohibitive requirements for both fitting times and memory requirements, since both requirements scale $O(Mnp^2)$, for M smoothing parameters and a model matrix with n rows and p columns (Wood et al., 2015). A beta location-scale model for only the island of Sardinia (24,000 km²; Appendix B) was interrupted after 144 hours, during which the model had only completed three steps using the Newton method described by Wood and Fasiolo (2017). In contrast, Gaussian models with fixed scaled parameters (i.e., constant variance) fit substantially faster and provided very similar spatialtemporal estimates of mean and variance in NDVI (Appendix B). Thus, we estimated trends in mean and variance in NDVI using a pair of GAMs for each region shown in Fig. A.4; one for estimating mean NDVI over space and time (assuming a constant variance), and one for estimating the mean squared residual (i.e., variance) in NDVI (again, assuming a constant scale parameter).

Together, the sets of models for mean NDVI (i.e., μ) and variance in NDVI (i.e., σ^2) had the following structure, with ν indicating NDVI:

$$\left\{ \begin{array}{l} \nu \sim \text{Normal}(\mu, \sigma^2) \\ \mu = \mathbb{E}(\nu) \approx \hat{\mathbb{E}}(\nu) = \hat{\mu} \\ \hat{\mu} = \beta_{\mu} + s(\text{prop_water}) + s(\text{elev_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) \\ \sigma^2 = \mathbb{E}((\nu - \mu)^2) \approx \hat{\mathbb{E}}((\nu - \hat{\mu})^2) = \hat{\sigma}^2 \\ \hat{\sigma}^2 = \beta_{\sigma^2} + s(\text{prop_water}) + s(\text{elev_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) + s(\hat{\mu}) \end{array} \right. , \quad (1)$$

where β_{μ} and β_{σ^2} are model intercepts, and, for a given (spatiotemporally aggregated) pixel: `prop_water` is the proportion of water within the pixel, `elev_m` is elevation above sea level in meters, `doy` is integer day of year (1-366), `year` is integer year (1981-2025), and `lat` and `long` are latitude and longitude, respectively. We omit the pixel-level indices for ease of readability. $s(\)$ indicates marginal smooth functions, while $t(\)$ indicates tensor product interaction terms. Seasonal terms used cyclical cubic splines (`bs = 'cc'`) with edge knots 0.5 and 366.5, while spatial smooths used a basis with second-order spline on the sphere (`bs = 'sos'`; Wood, 2017). The linear predictors for μ and σ^2 have the same structure, other than the $s(\mu)$ smooth term for σ^2 , but the functions for σ^2 are independent of those for μ . Despite the data aggregation procedures and faster fitting methods,

including parallelizing computation across 60 cores, each of the four models required multiple days to fit. Predicting from the models for $\hat{\mu}$ was also had substantial time and memory demands, since predictions were calculated without discrete prediction methods (`discrete = FALSE`). While discrete predictions were orders of magnitude faster, they were excessively spatially coarse. Predicting fitted values required over 27 days of fitting time despite consistently using > 1 TB of RAM. See Appendix C and the code availability statement for additional information.

From the two sets of models, we produced daily estimates of (1) the Dynamic Estimate of NDVI Variance (DENVar) and (2) the estimated mean NDVI they are conditional on, at a **XXX**° spatial scale. The rasters are freely available on GitHub (see code availability statement), along with long-term (1981-2025) averages of both DENVar and the estimated mean NDVI. Since there are no corrections to any bias of $\hat{\mathbb{E}}((\nu - \hat{\mu})^2)$, the estimates of σ^2 may be inflated near areas or periods of rapid and substantial change. Still, the appreciability of this inflation depends on the spatiotemporal scale over which changes are perceived (Levin, 1992; Frankham and Brook, 2004; Goodchild and Quattrochi, 2023; Mezzini et al., 2025). Since estimates of variance are conditional on the corresponding estimates of the mean, we suggest users always use the corresponding rasters of mean NDVI in their analyses rather than using independent estimates of mean NDVI. **DATA CAN BE DOWNLOADED VIA XXX.**

3 Results

The estimates of mean NDVI ($\hat{\mu}$) and DENVar ($\hat{\sigma}^2$) were spatially smooth but still captured large-scale trends (Fig. 2). The models for mean NDVI and DENVar explained **XXX%** and **XXX%** of the deviance, respectively. For both models, predictions were unreliable in areas with very scarce data (e.g., rock and ice biomes in the Arctic; see Fig. A.5), so such areas were excluded when producing figures and estimates of mean NDVI and DENVar.

PARAGRAPH DESCRIBING TRENDS IN MEAN AND VARIANCE OVER TIME (Fig. 3)

PARAGRAPH DESCRIBING TRENDS IN MEAN AND VARIANCE OVER TIME (Fig. 4).

Highly stochastic environments tended to have low: mean NDVI, human presence, species richness, annual area burned, annual precipitation, and But it's hard with these plots to be able to tell if it's just because of mean NDVI being low. Additionally, the plots use old versions of the rasters that included the excessively high values in the arctic.

PARAGRAPH DESCRIBING TRENDS IN MEAN AND VARIANCE OVER TIME (Fig. 5)

4 Discussion

We have presented a new spatiotemporally dynamic metric of environmental stochasticity, which we have called the Dynamic Estimate of NDVI Variance, DENVar. In this section, we discuss the interpretation of DENVar as well as the advantages and limitations of the statistical methods used.

4.1 Interpreting DENVar: the importance of scale

difficulties in estimating variance

what counts as a change? predictable vs stochastic changes

smoothness of mean impacts smoothness and size of variance: depends on scale of interest (e.g., animals' ability to respond to, learn, and predict changes in the mean). Changes not attributed to trends in the mean are stochastic

DENVar is a good metric as long as the spatiotemporal scale of interest is similar or larger to that of DENVar. May not work well for small-scale processes such as the movement dynamics of a small rodent, the phenology of a small orchard when spatiotemporal heterogeneity is finer than DENVar's resolution

4.2 Practical applications

Convert the subsections to a single subsection:

- spatial ecology
- movement behavior: (Mezzini et al., 2025)
- conservation (Marcus et al., 2025), etc.
- anthropology (navigating paper)
- industry (agriculture)

Environmental stochasticity and animal movement ecology

The field of animal movement ecology has long shown that animals alter their movement in response to resource abundance or availability (Burt, 1943; Southwood, 1977; Charnov, 1976; Broekman et al., 2024). More recently, movement ecologists have started focusing more on resource stochasticity (Stephens and Charnov, 1982; Rizzuto et al., 2021; Marcus and Noonan, 2023), but it is worth distinguishing between resource stochasticity and seasonality (Nilsen et al., 2005). Seasonality is generally best interpreted as a cyclical change in the mean that many animals can learn to predict, as in the case of green-wave surfing (Middleton et al., 2018; Geremia et al., 2019). In contrast, environmental stochasticity refers to unpredictable changes, such as forest fires [ref](#), sudden snow storms (Aikens et al., 2025), and unusual phenology [ref](#). Animals also respond to stochastic changes (Mezzini et al., 2025), but their ability to adapt depends on behavioral plasticity (Steixner-Kumar and Gläscher, 2020; Rickbeil et al., 2019), and memory (Polansky et al., 2015; Abrahms et al., 2019; Riotte-Lambert and Matthiopoulos, 2020; Falcón-Cortés et al., 2021; Ranc et al., 2022).

evolutionary tipping points (Botero et al., 2015)

multifarious change (Polazzo et al., 2024)

extreme events, black swans (Anderson et al., 2017; Logares and Nuñez, 2012)

- migration routes are key habitat (Ortega et al., in prep), especially in stochastic environments

- 213 • can fail to adapt to changes in conditions (Sawyer et al., 2019)
- 214 • <https://www.sciencedirect.com/science/article/pii/S0960982220308484>
- 215 • <https://onlinelibrary.wiley.com/doi/full/10.1111/gcb.15169>
- 216 • industrial development affects animals' ability to respond to both predictable and unpredictable
- 217 change (<https://www.nature.com/articles/s41559-022-01887-9>)
- 218 • effects of sudden deep snow in red desert on proghorn (<https://www.sciencedirect.com/science/article/abs/pii/S0960982220308484>)
- 219 • <https://www.nature.com/articles/s41467-023-37750-z>
- 220 • <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.4238>
- 221 • detecting regime shifts (in lakes: (Bjorndahl et al., 2022))

222 **Environmental stochasticity and population dynamics**

223 read lande_risks_1993

224 Environmental stochasticity reduces habitats' energetic balance and carrying capacity (Chevin et al.,
 225 2010) as individuals struggle to rely on unpredictable resources and increase the instability of pop-
 226 ulations and communities (Lande, 1993). Additionally, stochastic environments reduce the ability
 227 of specialist species to adapt and evolve while selecting for more generalist species (Levins, 1974).
 228 Consequently, stochastic environments tend to have lower species diversity (Marcus et al., 2025).
 229 Recent work by Devries has suggested that NDVI stochasticity is an important metric for evaluating
 230 forest recovery after large fires, while Collinson has suggested that variance in NDVI is an important
 231 predictor of the occurrence of lightning-induced forest fires.

- 232 • jonzen_rapid_2006
- 233 • effects can be compounded by cumulative effects (e.g., loss of species diversity; (Wolf et al.,
 234 2017))
- 235 • climate change affects plant phenology, but changes in situ may not match experimental esti-
 236 mates (Wolkovich et al., 2012)
- 237 • reproduction of painted lady butterfly (Stefanescu et al., 2021)
- 238 • wessling_seasonal_2018

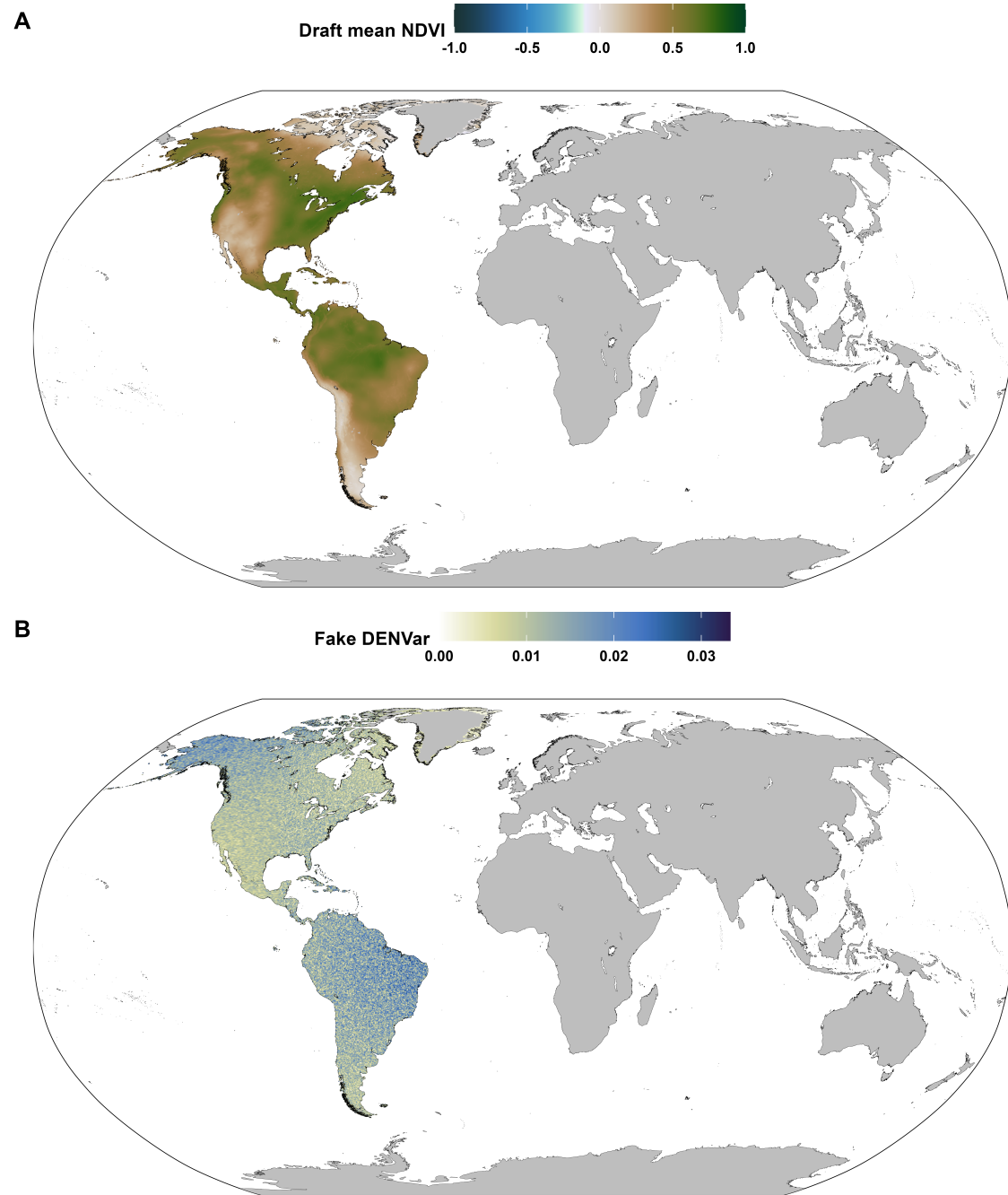


Figure 2: Estimated long-term mean NDVI (A) and DENVar (B).

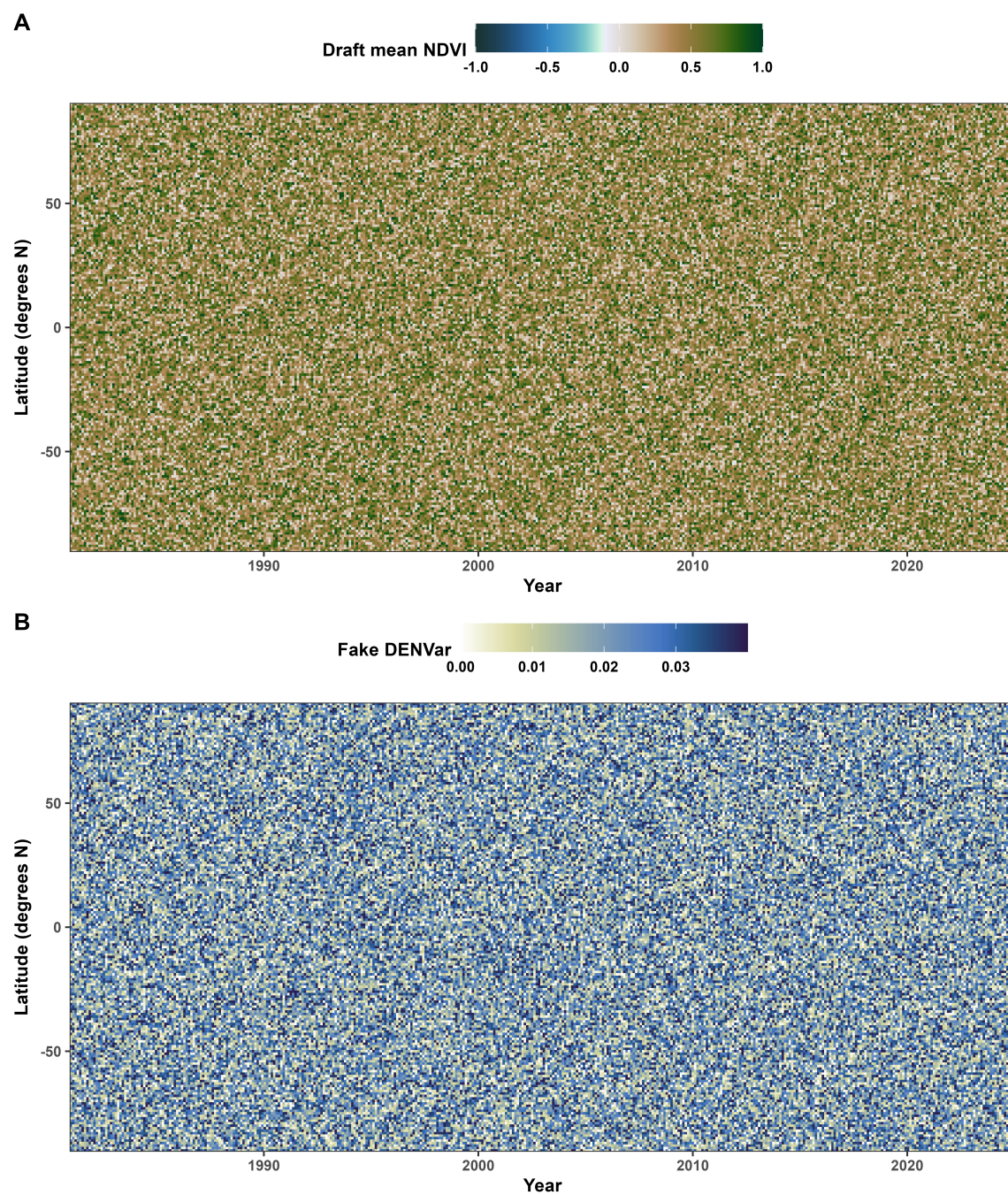


Figure 3: Surface plots of mean NDVI (A) and DENVar (B) over time and latitudes.

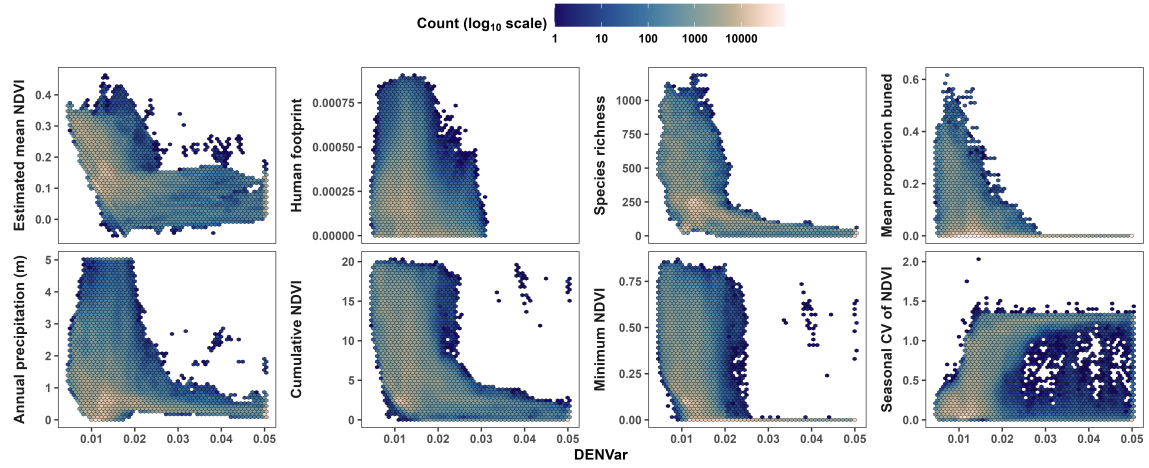


Figure 4: Hex plots of the relationship between the long-term global Dynamic Estimate of NDVI Variance (DENVar) and common ecological metrics. The fill of the hexagonal cells represents the log-10 number of points within the cell. Rowwise, the variables are: long-term mean NDVI (as estimated in this manuscript), the machine-learning Human Footprint Index (ml-HFI) of , and the 2015 cumulative, minimum, and seasonal range in the Dynamic Habitat Index of (Radeloff et al., 2019).

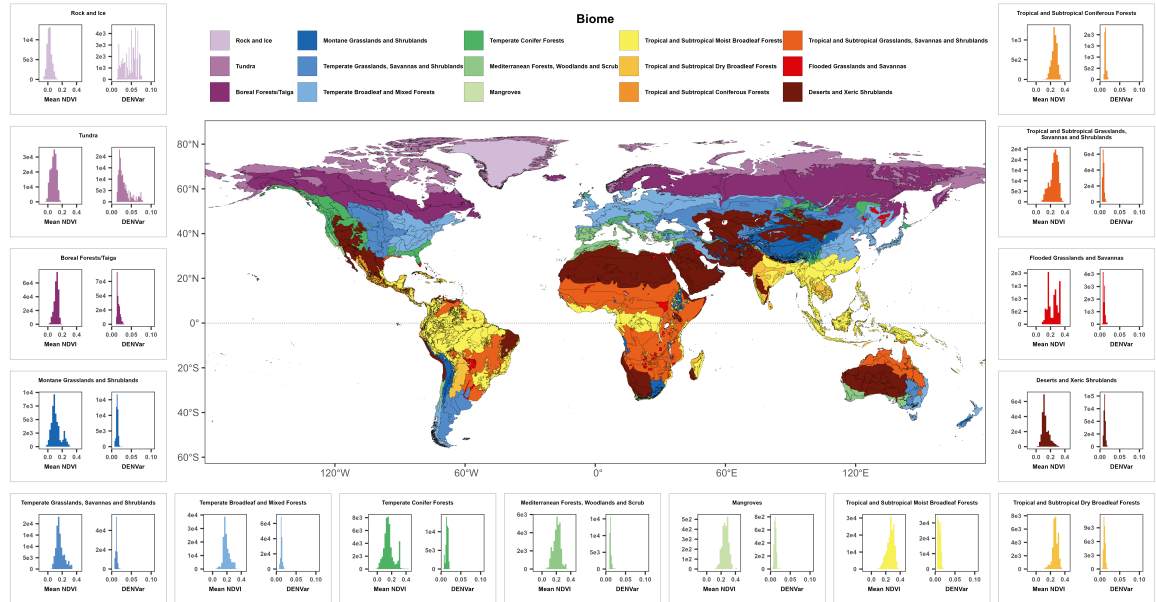


Figure 5: Map of terrestrial biomes along with hex plots of mean NDVI and DENVar for each biome. **UPDATE FIGURE TO HEXPLOTS.**

5 Other useful references

- **READ**: Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated Using Simulated Ecological Data (<https://doi.org/10.1371/journal.pone.0041010>)
- **READ**: <https://doi.org/10.1029/2024GL109298>
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- rickbeil_plasticity_2019
- mueller_search_2008
- pettorelli_using_2005
- nilsen_can_2005
- merkle_large_2016
- tian_evaluating_2015
- huang_commentary_2021
- fan_global_2016
- wang_stochastic_2019
- pease_ecological_2024
- xu_plasticity_2021
- calibrating across sensors: <https://ieeexplore.ieee.org/abstract/document/6809983>

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267 The authors would like to thank Dr. Simon Wood for writing the custom Beta location-scale family
268 even though it was not used in the final models.

269 **Author Contributions**

270 **Funding**

271 **Conflicts of Interest**

272 The authors declare that there is no conflict of interest regarding the publication of this article.

273 **Code and Data Availability**

274 All code is available on GitHub at <https://github.com/QuantitativeEcologyLab/ndvi-stochasticity>,
275 including code for downloading the original NDVI rasters directly from the servers they are housed
276 in. Most data are not included due to GitHub's size limitations, but all objects can be created using
277 the code provided.

278 **Supplementary Materials**

- 279 • Appendix A: Color palettes used and input data.
- 280 • Appendix B: Sensitivity analyses.
- 281 • Appendix C: Code for fitting the models, along with supporting supplementary figures.

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